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R.V.COLLEGE OF ENGINEERING,BENGALURU-560059

(Autonomous institution affiliated to VTU,Belagavi)

Department of Electronics and Communication Engineering

Semester: III

Test-I

Course Title: Analysis and Design of Digital Circuits **Course Code: 16EC34**

Duration: $1\frac{1}{2}$ *Hrs* **Date:**

Maximum Marks:50

Sl.No	QUESTION	M	BT	CO
1.a	Simplify the Boolean expression Y= ∑m(1,2,3,7,9,11,17,18,23,25,29,31) using K-map method.	5	L3	CO1
1.b	Design a 4-bit carry look ahead adder with the help of relevant Boolean expressions.	5	L2	CO1
2	Design an application specific decoder to drive common cathode seven segment display.	10	L4	CO4
3	Design a 3-bit *X* 2-bit binary multiplier using suitable logic devices.	10	L3	CO3
4.a	Realize the Boolean expression F= xy'+ y'z using suitable multiplexer.	5	L2	CO1
4.b	Design a 6-bit magnitude comparator using 7485 IC's.	5	L3	CO2
5.a	Design an exces-3 to binary code converter using suitable parallel adder.	5	L4	CO3
5.b	Design a gated SR-latch using NAND gates. Write the function table and derive the characteristic equation for the same.	5	L2	CO1

BT-Blooms Taxonomy,CO-Course Outcomes,M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	20	5	15	10	0	15	20	15	-	-

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